Course unit catalogue

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1 First year

1.1 MATH10131 - Calculus and Vectors B

Credit Rating: 15

Aims. The course unit unit aims to provide an introduction to the basic elements of calculus.

Overview. This lecture course introduces the basic ideas of complex numbers relating them the standard transcendental functions of calculus. The basic ideas of the differential and integral calculus are revised and developed. Vectors in two and three dimensions are introduced and this leads on to the calculus of functions of more than one variable and the beginnings of vector calculus.

Learning outcomes. On successful completion of this module students will have acquired an active knowledge and understanding of some basic concepts and results in calculus.

Syllabus.

- **Numbers and Functions.** Basic algebra of real and complex numbers; real line and complex plane; graphs and curve sketching; functions, domain and range, inverse functions; standard functions and inverse functions; basic algebra of real and complex numbers.
- **Limits and Differentiation.** Basic notion of limit and continuity; discontinuities, left and right limits; finding some limits; definition of derivative; derivatives of standard functions and their inverses; sums, products, quotients and the chain rule; implicit functions; logarithmic differentiation; higher derivatives (use in curve sketching).
- **Infinite Series.** Notation, basic notions of convergence, radius of convergence; infinite Taylor s series; expansions for standard functions.
- **More on Complex Numbers.** Euler s Theorem and De Moivre s Theorem; polar form of complex numbers (polar representation of the plane); roots of unity; complex forms of sin and cos, relationship to trigonometric identities.
- **Integration.** Definite and indefinite integrals; Fundamental Theorem of Calculus; techniques: linearity, integration by parts, partial fractions, substitution; lengths of curves, surfaces and volumes of revolution.
- **Vectors in 2 and 3 Dimensions.** Representation as directed line segments (magnitude, direction); choice of axes, components, Cartesian representation; basic properties, addition, subtraction, polar representation and relationship with complex numbers in 2 dimensions; scalar and vector product; representation of lines, planes, curves and surfaces.
- **Functions of more than One Variable.** Partial derivative, chain-rule, Taylor expansion; turning points (maxima, minima, saddle-points); grad, div and curl and some useful identities in vector calculus; integration in the plane, change of order of integration; Jacobians and change of variable; line integrals in the plane; path-dependence, path independence; Stokes theorem and Green s theorem.

1.2 MATH10232 – Calculus and Applications B

Credit Rating: 15

Aims. This course unit aims to introduce students to ordinary differential equations, primarily covering methods of solution and applications to physical situations.

Overview. The unit will cover first and second order ordinary differential equations including classification and standard solution methods. Applications will be drawn from the field of classical mechanics, but no prior experience in mechanics is expected or required. Matlab will be used to illustrate some of the ideas and methods.

Learning outcomes.

- On completion of this unit successful students will be able to solve first order and second order linear problems and first order separable equations analytically.
- Use substitution methods and power series methods to find solutions.
- Be able to investigate solutions using direction fields and Euler s method.
- Have used Matlab as a mathematical tool and used differential equations to solve problems in mechanics and other applications.

Syllabus.

1. **Introduction:** concept of mathematical modeling; definition and classification of ordinary differential equations; order; linear and autonomous equations.
2. **First-order ordinary differential equations:** separable, graphical and numerical solutions; initial conditions; direction fields; Euler s method; use of Matlab; existence and uniqueness of solutions; integrating-factor methods; power series; substitution methods for nonlinear equations; phase plane and stability; population modelling.
3. **Higher-order ordinary differential equations:** general solution of linear, second-order equations; initial and boundary conditions; homogeneous equations; particular integrals; method of undetermined coefficients; harmonic oscillators; resonance; coupled systems; phase portraits; substitution methods for nonlinear equations; plane autonomous systems; predator-prey systems.
4. **Mechanics:** particle kinematics in Cartesian and polar coordinates; Newton s laws; forces; Newton s law of gravitation; work and energy; motion along a line (potential wells); equilibrium and stability; simple systems of particles; simple pendulum; compound pendulum; double pendulum.

1.3 MATH10212 – Linear Algebra B

Credit Rating: 15

Aims. This course unit aims to introduce the basic ideas and techniques of linear algebra for use in many other lecture courses. The course will also introduce some basic ideas of abstract algebra and techniques of proof which will be useful for future courses in pure mathematics.

Overview. This core course aims at introducing students to the fundamental concepts of linear algebra culminating in abstract vector spaces and linear transformations. The course starts with systems of linear equations and some basic concepts of the theory of vector spaces in the concrete setting of real linear n-space, Rn. The course then goes on to introduce abstract vector spaces over arbitrary fields and linear transformations, matrices, matrix algebra, similarity of matrices, eigenvalues and eigenvectors. The subject material is of vital importance in all fields of mathematics and in science in general.

Learning outcomes.

On successful completion of this course unit students will be able to:

- solve systems of linear equations by using Gaussian elimination to reduce the augmented matrix to row echelon form or to reduced row echelon form;
- understand the basic ideas of vector algebra: linear dependence and independence and spanning;
- be able to apply the basic techniques of matrix algebra, including finding the inverse of an invertible matrix using Gauss-Jordan elimination;
- know how to find the row space, column space and null space of a matrix, and be familiar with the concepts of dimension of a subspace and the rank and nullity of a matrix, and to understand the relationship of these concepts to associated systems of linear equations;
- be able to find the eigenvalues and eigenvectors of a square matrix using the characteristic polynomial and will know how to diagonalize a matrix when this is possible;
- be able to recognize and invert orthogonal matrices;
- be able to orthogonally diagonalize symmetric matrices;
- be able to find the change-of-basis matrix with respect to two bases of a vector space;
- be familiar with the notion of a linear transformation and its matrix.

Future topics requiring this course unit. Almost all Mathematics course units.

Syllabus.

- **Linear Equations in Linear Algebra:** Systems of Linear Equations - Row Reduction and Echelon Forms - Vector Equations - The Matrix Equation Ax=b - Solution Sets of Linear Systems - Applications of Linear Systems - Linear Independence - Introduction to Linear Transformations - The Matrix of a Linear Transformation [Lay, Chapter 1, 6 lectures]
- **Matrix Algebra:** Matrix Operations - The Inverse of a Matrix - Characterizations of Invertible Matrices - Partitioned Matrices - Matrix Factorizations - Subspaces of Rn - Dimensions and Rank [Lay, Chapter 2, 4 lectures]
- **Determinants:** Introduction to Determinants - Properties of Determinants - Cramer s Rule, Volume, and Linear Transformations [Lay, Chapter 3, 4 lectures]
- **Vector Spaces:** Vector Spaces and Subspaces - Null Spaces, Column Spaces, and Linear Transformations - Linearly Independent Sets; Bases - Coordinate Systems - The Dimension of Vector Space Rank - Change of Basis [Lay, Chapter 4, 6 lectures]

- **Eigenvalues and Eigenvectors:** Eigenvectors and Eigenvalues - The Characteristic Equation Diagonalization - Eigenvectors and Linear Transformations - Complex Eigenvalues [Lay, Chapter 5, 6 lectures]
- **Orthogonality:** Inner Product, Length, and Orthogonality - Orthogonal Sets - Orthogonal Projections - The Gram- Schmidt Process - Inner Product Spaces - Applications of Inner Product Spaces [Lay, Chapter 6, 4 lectures]
- **Symmetric Matrices:** Diagonalization of Symmetric Matrices [Lay, Chapter 7, 2 lectures]

1.4 MATH10111 – Sets, Numbers and Functions B

Credit Rating: 15

Aims. The aims of this course are to provide a basic introduction to various methods of proof used in mathematics and to the fundamental ideas in the study of sets, numbers and functions.

Overview. This lecture course is intended to introduce students to the concept of "proof". The objects of study, sets, numbers and functions, are basic to almost all Mathematics.

Learning outcomes.

On successful completion of this module students will be:

- familiar with and able to manipulate the basic concepts of Pure Mathematics such as sets and functions;
- able to construct elementary proofs of mathematical statements utilizing inductive arguments and arguments by contradiction;
- able to understand proofs of such results at the Fundamental Theorem of Arithmetic and the Euclidean Algorithm;
- familiar with the definitions and know some examples of groups and fields.

Syllabus.

- **The language of mathematics.** Mathematical statements, quantifiers, truth tables, proof.
- **Number theory I.** Prime numbers, proof by contradiction
- **Proof by induction.** Method and examples.
- **Set Theory.** Sets, subsets, well known sets such as the integers, rational numbers, real numbers.Set Theoretic constructions such as unions, intersections, power sets, Cartesian products.
- **Functions.** Definition of functions, examples, 1-1 and onto functions, bijective functions, composition of functions, inverse functions.
- **Counting.** Counting of (mostly) finite sets, inclusion-exclusion principle, pigeonhole principle, binomial theorem.
- **Euclidean Algorithm.** Greatest common divisor,proof of the Euclidean Algorithm and some consequences, using the Algorithm.

- **Congruence of Integers.** Arithmetic properties of congruences,solving certain equations in integers.
- **Relations.** Examples of various relations,reflexive, symmetric and transitive relations. Equivalence relations and equivalence classes. Partitions.
- **Number Theory II.** Fundamental theorem of Arithmetic, Fermat s little theorem.
- **Binary Operations.** Definition and examples of binary operations. Definition of groups and fields with examples. Proving that integers mod p (p a prime) give a finite field.

2 Second year

2.1 MATH20122 – Metric Spaces

Credit Rating: 10

Aims. The course unit aims to introduce the basic ideas of metric spaces.

Overview. A metric space is a set together with a good definition of the distance between each pair of points in the set. Metric spaces occur naturally in many parts of mathematics, including geometry, fractal geometry, topology, functional analysis and number theory. This lecture course will present the basic ideas of the theory, and illustrate them with a wealth of examples and applications. This course unit is strongly recommended to all students who intend to study pure mathematics and is relevant to all course units involving advanced calculus or topology.

Learning outcomes.

On completion of this unit successful students will be able to:

- deal with various examples of metric spaces;
- have some familiarity with continuous maps;
- work with compact sets in Euclidean space;
- work with completeness;
- apply the ideas of metric spaces to other areas of mathematics.

Syllabus.

1. **Basic Definitions.** Euclidean metric, taxicab metric, discrete metric, edge metric, word metric, sup metric, L1 metric, Hausdorff metric, l2 metric, product metrics. Examples. [4 lectures]
2. **Open and Closed Sets.** Interior, closure, sequences and convergence, boundary. Denseness. Equivalent metrics. Examples. [4]
3. **Uniform Convergence.** Sequences of continuous functions. Examples. [2]
4. **Continuous maps.**

   Extending the elementary definition. Relationship with open sets, sequences. Examples [4]

5. **Compactness.** Open coverings. Continuous maps on compact sets. Compactness in Euclidean space. [4]

6. **Completeness.** Cauchy sequences. The Contraction Mapping Theorem, Examples. [3]

2.2 MATH20111 – Real Analysis

Credit Rating: 10

Aims. The course unit unit aims to introduce the basic concepts of limit and convergence (of real sequences, series and functions) and to indicate how these are treated rigorously, and then show how these ideas are used in the development of real analysis.

Overview. The first part of the course discusses the convergence of real sequences and series. The second part of the course discusses the concept of limit for real-valued functions of a real variable. This concept is then used to define and investigate the concepts of continuity and differentiability for such functions.

Learning outcomes.

On completion of this unit successful students will be able to:

- evaluate the limits of a wide class of real sequences;
- determine whether or not real series are convergent by comparison with standard series or using the Ratio Test;
- understand the concept of continuity and be familiar with the statements and some proofs of the standard results about continuous real functions;
- understand the concept of the differentiability of a real valued function and be familiar with the statements of the standard results about differentiable real functions.

Syllabus.

- **Sequences.** Null sequences and the standard list of null sequences. Convergent sequences, the Algebra of Limits, divergent sequences, monotone bounded sequences
- **Series.** Convergent and divergent series, geometric series and the harmonic series. Series with non-negative terms, the Comparison Test, the Limit Comparison Test, the Ratio Test.
- **Continuity.** Limits of real functions, sums, products and quotients of limits. Continuity of real functions, sums, products and quotients of continuous functions, the composition of continuous functions. The standard results about continuous real functions: the Intermediate Value Theorem and the Boundedness Theorem.
- **Differentiability.** Differentiability of real-valued functions, sums, products and quotients of differentiable functions, Rolle s Theorem, the Mean Value Theorem, Cauchy s Mean Value Theorem.

2.3 MATH20201 – Algebraic Structures 1

Credit Rating: 10

Aims. The course unit unit aims to introduce basic ideas group theory with a good range of examples so that the student has some familiarity with the fundamental concepts of abstract algebra and a good grounding for further study.

Overview. This course unit provides an introduction to groups, one of the most important algebraic structures. It gives the main definitions, some basic results and a wide range of examples. This builds on the study of topics such as properties of the integers, modular arithmetic, and permutations included in MATH10101/MATH10111. Groups are a fundamental concept in mathematics, particularly in the study of symmetry and of number theory.

Learning outcomes.

On completion of this unit successful students will be able to:

- Appreciate and use the basic definitions and properties of groups;
- Command a good understanding of the basic properties for a good range of examples;
- Understand and find simple proofs of results in group theory.

Syllabus.

- **Binary operations.** Multiplication tables, associativity, commutativity, associative powers. [2 lectures]
- **Groups.** Definitions and examples (groups of numbers, the integers modulo n, symmetric groups, groups of matrices). [2]
- **Subgroups.** Subgroup criterion, cyclic subgroups, centralizer, centre, order of an element. [4]
- **Cyclic groups.** Subgroups of cyclic groups are cyclic, subgroups of finite cyclic groups. [1]
- **Cosets and Lagrange s Theorem.** [2]
- **Homomorphisms and isomorphisms.** Definition and examples, group theoretic properties. [2]
- **Conjugacy.** Conjugacy classes, conjugacy in symmetric groups, the class formula. [4]
- **Normal subgroups.** [2]
- **Factor groups.** [2]
- **The First Isomorphism Theorem.** [1]

2.4 MATH20142 – Complex Analysis

Credit Rating: 10

Aims. The course unit unit aims to introduce the basic ideas of complex analysis, with particular emphasis on Cauchy s Theorem a nd the calculus of residues.

Overview. This course introduces the calculus of complex functions of a complex variable. It turns out that complex differentiability is a very strong condition and differentiable functions behave very well. Integration is along paths in the complex plane. The central result of this spectacularly beautiful part of mathematics is Cauchy s Theorem guaranteeing that certain integrals along closed paths are zero. This striking result leads to useful techniques for evaluating real integrals based on the calculus of residues .

Learning outcomes.

On completion of this unit successful students will be able to:

- understand the significance of differentiability for complex functions and be familiar with the Cauchy-Riemann equations;
- evaluate integrals along a path in the complex plane and understand the statement of Cauchy s Theorem;
- compute the Taylor and Laurent expansions of simple functions, determining the nature of the singularities and calculating residues;
- use the Cauchy Residue Theorem to evaluate integrals and sum series.

Syllabus.

1. **Series.** Complex series, power series and the radius of convergence. [2 lectures]
2. **Continuity.** Continuity of complex functions [2]
3. **The complex plane.** The topology of the complex plane, open sets, paths and continuous functions. [2]
4. **Differentiation.** Differentiable complex functions and the Cauchy-Riemann equations. [2]
5. **Integration.** Integration along paths, the Fundamental Theorem of Calculus, the Estimation Lemma, statement of Cauchy s Theorem. [4]
6. **Argument and Logarithm.** [2]
7. **Taylor and Laurent Series.** Cauchy s Integral Formula and Taylor Series, Zeros and Poles, Laurent Series. [3]
8. **Residues.** Cauchy s Residue Theorem, the evaluation of definite integrals and summation of series. [5]

2.5 MATH20411 – Partial Differential Equations and Vector Calculus B

Credit Rating: 10

Aims. This course introduces students to analytical and numerical methods for solving partial differential equations (PDEs) and builds on the first year core applied mathematics courses to develop more advanced ideas in differential and integral calculus.

Overview. The main topics to be explored are: Fourier series, partial differential equations, analytical and numerical methods for solving classical PDEs (Laplace s equation and the heat and wave equations) and several topics in vector calculus, including surface and volume integrals. The course covers similar material to MATH20401 but contains a reduced range of topics and with fewer details, where appropriate. The methods employed in the course will prove essential for all of the applied mathematics and numerical analysis options in the remaining semesters of the Joint Honours BSc and MMath degree programmes.

Learning outcomes.

On completion of this unit successful students will be able to:

- convert Cartesian coordinates into cylindrical and spherical coordinates and sketch surfaces expressed in all of these coordinate systems;
- perform partial differentiation of functions of more than one variable and interpret these derivatives physically;
- recognise classical PDEs describing physical processes such as heat diffusion;
- interpret boundary and initial conditions physically;
- understand the basic concept of orthogonal functions;
- compute Fourier series, Fourier sine series and Fourier cosine series of some piecewise continuous functions;
- classify second-order PDEs as being either elliptic, hyperbolic or parabolic;
- solve, analytically, via the method of separation of variables, the heat and wave equations (in one space variable) and Laplace s equation (in two space variables), on rectangular and circular domains;
- solve, numerically, via finite difference schemes, the heat equation in one space variable;
- solve, numerically, convection-diffusion equations using upwind finite differencing;
- compute elements of surface and volume in different coordinate systems;
- evaluate line, surface and volume integrals over a range of domains, using transformations to other coordinate systems where appropriate;
- understand grad, div and curl operator notation and relate some key identities to physical properties of vector fields;
- understand, and interpret physically, the classical Divergence, Green s and Stokes theorems.

Syllabus.

- **Introductory material.** [3 lectures] Cartesian, cylindrical and spherical coordinates. Functions of several variables, surfaces. Partial derivatives, chain rule. Partial differential equations, boundary and initial conditions. Integrals of functions of several variables.

- **Fourier series.** [4 lectures] Orthogonality. Fourier series and Fourier coefficients. Periodic, even and odd functions. Fourier s theorem. Fourier sine and cosine series.
- **Partial Differential Equations.** [2 lectures] Linearity, homogeneity and order of PDEs. Classification of second-order equations. Introduction to the classical equations: Laplace s, heat and wave equations.
- **Analytical Solution of PDEs.** [4 lectures] The method of separation of variables. Solving, exactly, initial-value problems for the heat and wave equations. Solving Laplace s equation in both Cartesian and plane polar co-ordinates.
- **Numerical Solution of PDEs.** [4 lectures] Solving, approximately, the reaction-diffusion and convection-diffusion equations (ODEs) via finite difference methods. Solving, approximately, the heat equation in one space variable (PDE). Explicit and implicit numerical schemes.
- **Vector Calculus.** [5 lectures] Surfaces, unit vectors, elements of surface/volume. Line, surface and volume integrals. Scalar and vector fields: differential and integral calculus. Grad, div and curl operators and related identities. Classical theorems: Divergence, Stokes theorems.

2.6 MATH20212 – Algebraic Structures 2

Credit Rating: 10

Aims. The course unit unit aims to introduce the algebraic structures of rings and fields; describe the quotient structure and its connection with homomorphisms of rings; present important examples rings and develop some of their properties with particular emphasis on polynomial rings and factorisation in rings.

Overview. This course builds on Algebraic Structures 1, which is a prerequisite, and continues the strong emphasis on examples. The algebraic structures of rings and fields will be introduced. The construction of quotient rings and the relationship with homomorphisms is one of the main themes. These ideas will be used to construct roots of polynomials in extension fields. Factorisation in polynomial rings and rings of integers of number fields will also be studied.

Learning outcomes.

On completion of this unit successful students will be able to:

- demonstrate their knowledge of the definition of a ring and important examples;
- demonstrate their understanding of the quotient construction in theoretical terms and also in particular contexts;
- demonstrate their understanding of how to produce roots of polynomials in extension fields and be able to compute in such fields;
- solve a range of problems which require understanding of rings and fields;
- apply theoretical results to computations in particular examples of rings.

Syllabus.

1. **Definitions and examples of rings** (rings of numbers, rings of matrices, quaternions, rings of endomorphisms, group rings, rings of polynomials, subrings); [4 lectures]
2. **Domains, fields and division rings;**

   nilpotent and idempotent elements, products of rings; (many) examples; with students gaining familiarity with the ideas and examples through attempting exercises. [4]
3. **Isomorphisms and homomorphisms (of rings):** what is preserved and reflected; kernel of a homomorphism, ideals; principal ideals, operations on ideals. [4]
4. **The quotient construction (for rings):** the construction and connection with homomorphisms; maximal ideals; ideals of the quotient ring; examples. [3]
5. **Polynomial rings and unique factorisation:** polynomial rings; division algorithm; unique factorisation. [3]
6. **Constructing roots of polynomials:** construction of extension fields; examples, including finite fields. [4]

3 Third year

3.1 MATH20902 – Discrete Mathematics

Credit Rating: 10

Aims. This module aims to engage students with a circle of algorithmic techniques and concrete problems arising in elementary number theory and graph theory.

Overview. Modern Discrete Mathematics is a broad subject bearing on everything from logic to logistics. Roughly speaking, it is a part of mathematics that touches on those subjects that Calculus and Algebra can t: problems where there is no sensible notion continuity or smoothness and little algebraic structure. The subject, which is typically concerned with finiteor at the most countable-sets of objects, abounds with interesting, concrete problems and entertaining examples.

Learning outcomes.

Students should develop the ability to think and argue algorithmically, mainly by studying examples and applications in graph theory and combinatorics. On completion of this unit successful students will be able to:

- understand proofs by induction thoroughly and be fluent in their construction;
- read and understand written descriptions of algorithms;
- develop and apply simple algorithms to solve problems or prove theorems;
- give the basic definitions of graph theory and a range of standard examples including, for example, complete graphs and bipartite graphs;
- characterise planar graphs and prove the five-colour theorem.

Syllabus.

- **Graph Theory & Applications:** [22] The basic definitions about graphs should be familiar from the Mathematical Workshop MATH10001, so after a brief review we will treat the following topics:
  - Basic notions & notations, trees [5]
  - Eulerian tours & Hamiltonian cycles [4]
  - The Principle of Inclusion/Exclusion and the Matrix-Tree Theorem [4]
  - Shortest paths & applications to scheduling [4]
  - Planar graphs & map colouring [5]

3.2 MATH36032 – Problem Solving by Computer

Credit Rating: 10

Aims. To develop skills in translating mathematical ideas into MATLAB programs, thereby using the computer as a tool to investigate and solve mathematical problems.

Overview. This module is concerned with using a modern computer software package for solving mathematical problems and hence it involves a reasonable amount of computer programming. The student will be given a thorough introduction to the capabilities of the state of the art software package MATLAB, covering numeric, symbolic (with the Symbolic Math Toolbox) and graphical features. Although MATLAB is the chosen course software, the emphasis will be given to principles that are not specific to any particular package. Basic principles of technical writing will also be taught, and the student will apply these to the written projects, which must be produced using a wordprocessing/typesetting package.

Learning outcomes.

- On successful completion of the course unit students will be able to express the solution of a number of projects, using a combination of symbolic, numeric and graphical forms, in terms of robust and elegant programs employing the software package MATLAB.
- The projects, different each year, will be chosen from a variety of mathematical topics, including linear algebra, differential equations, multivariate calculus, mathematical finance, statistics, dynamical systems, and nonlinear equations, often linked to particular practical applications.
- Students will also have gained knowledge and experience of technical writing.

Future topics requiring this course unit. None.

Syllabus.

1. **General Introduction:** Brief history of mathematical computing. Mathematical software packages, programming languages.
2. **Programming in MATLAB:** Essentials of MATLAB; vectors and matrices, colon notation, numeric output, graphics, control structures and logical tests. MATLAB functions. Symbolic and high precision computations.

3. **Projects:** Three projects will be set on mathematical topics, often with applications. No special background knowledge is required and the relevant theory will be covered in lectures. Marks will be awarded for mathematical content and correctness, use of MATLAB, and technical writing and presentation.

4. **Laboratory Work:** Approximately half of the total of 36 contact hours will be spent on lectures and the remainder on supervised computer laboratory work on the projects. Students should expect to spend more than the allocated class-time on the computer to produce their projects.

3.3 MATH20302 – Introduction to Logic

Credit Rating: 10

Aims. The course unit unit aims to introduce the student to the idea of formalising arguments, both semantically and syntactically, and to the fundamental connection between these approaches.

Overview. Logic is the study of arguments: what they are and what it means to say that they are sound. As such it is central to Mathematics, Philosophy and, to an increasing extent in recent years, Computer Science. Most of this course unit will deal with the most basic sort of logical argument (i.e., in everyday parlance, what we mean by A follows from B), namely those which depend for their soundness simply on the commonly agreed interpretations of the logical connectives not , and , or and implies . That is referred to as propositional logic. We shall characterise the above notion of follows in two fundamentally different ways, firstly in terms of preservation of truth (semantically), and secondly in terms of the formal rules it obeys (proof theoretically, or syntactically). The highlight of the course unit will be the Completeness Theorem for Propositional Logic, which tells us that these two quite different characterisations are equivalent. This is the younger sibling of the Completeness Theorem for Predicate Logic.That is a fundamental result for Mathematics which says that one can set up a notion of formal mathematical proof which is strong enough to capture mathematical truth: if something isn't formally provable then there must exist a counterexample. The propositional logic version of that theorem already allows us to introduce the key concepts and to see how a proof of such a general result can be achieved. In the final few lectures the student is introduced to some basic ideas about predicate logic. This deals with languages which allow the use of variables together with the quantifiers "for all..." and "there exists..." as well as functions, relations and constants. Such languages are of far greater expressive power than those of propositional logic and using them one can, in principle, formalise any mathematical argument. The Compactness Theorem for Predicate Logic will be stated and simple applications will be given. This part of the course unit serves as a brief introduction to these logics with quantifiers and to concepts which will be used in some level 3/4 units in Logi (for instance, the level 3 logic course contains a proof of the Compactness Theorem).

Learning outcomes.

On completion of this unit successful students will be able to:

- formalise arguments in propositional logic both semantically and syntactically and understand how these are connected (via the Completeness Theorem);
- in simple cases be able to determine whether A follows from B is true, using a variety of methods;

- be able to interpret formulas and sentences of predicate logic in mathematical structures and apply the compactness theorem in simple cases.

Future topics requiring this course unit. The course unit forms a coherent subject on its own, and provides necessary background knowledge for the third and fourth level Logic course units.

Syllabus.

1. Motivation, syntax, propositional variables, connectives, propositional terms. [3]
2. Valuations, logical consequence, logical equivalence, truth tables, satisfiability, disjunctive normal form, interpolation, adequate sets of connectives. [5]
3. Rules of proof, formal proofs, the Soundness Theorem. [4]
4. The Completeness and Compactness Theorems. [4]
5. Languages for Predicate Logic, structures, interpretations, and the statement of the Compactness Theorem for Predicate Logic, with examples. [6]

3.4 MATH32072 - Number Theory

Credit Rating: 10

Aims. This lecture course aims to introduce students to some elementary concepts in number theory. We shall apply some basic algebraic concepts in order to study integer solutions of certain polynomial equations with integer coefficients, such as $x^2 - dy^2 = 1$ (Pell's equation). Whilst there are no official prerequisites for this course, please note that: Students who already have some familiarity with groups and other basic algebraic structures may find the material slightly easier than those students who don't. A certain level of mathematical maturity is required and you will be expected to work hard!

Overview. Number theory is arguably one of the oldest and most fascinating branches of mathematics. This fascination stems from the fact that there are a great many theorems concerning the integers, which are extremely simple to state, but turn out to be rather hard to prove. Even more tantalisingly, there are many simple questions that one can ask about the integers, to which no answer is yet known. It may come as no surprise to you that many of these theorems and open problems relate to the mysterious nature of prime numbers. In this course we will use a few elementary ideas along with one or two algebraic concepts to prove some appealing statements such as every positive integer is the sum of four squares and there are infinitely many primes of the form 4n+1. We will conclude the course by mentioning a few interesting open problems.

Learning outcomes.

On successful completion of this course unit students will:

- Be able to describe and use the Euclidean algorithm and to explain how to solve linear Diophantine equations.
- Be able to describe and use the continued fraction algorithm to (a) find representations of rationals and quadratic irrationals, (b) obtain rational approximations to real numbers and (c) find solutions to Pell's equations.

- Be able to describe the group of units modulo n for fixed value of n, giving information on its group structure and in particular determining the existence of a primitive root modulo n.
- Be able to state and prove several theorems involving polynomial congruences.
- Be able to describe and use the theorem of quadratic reciprocity (and other results) to work with Legendre symbols.

Syllabus.

- **Division, primes and the Euclidean algorithm [2 lectures]**
  - Introduction to the course and discussion of puzzles and problems involving Diophantine equations.
  - A review of basic notions (divisibility, primes, greatest common divisor).
- **Finite continued fractions [3 lectures]**
  - Finite simple continued fractions and the continued fraction algorithm.
  - Connection to the Euclidean algorithm.
- **Infinite simple continued fractions [4 lectures]**
  - Definitions and convergence properties.
  - Representation of real numbers; finite/periodic/purely periodic continued fractions.
  - Rational approximations of real numbers.
- **Pell s equations [3 lectures]**
  - Definitions and examples; trivial solutions and generating positive solutions.
  - Using continued fractions to find solutions.
- **The group of units modulo n [2 lectures]**
  - A review of basic concepts (modular arithmetic, groups, units) with examples.
  - Wilson s theorem, Euler s theorem and Fermat s little theorem
  - An application of Euler s theorem (RSA)
- **Polynomial congruences and primitive roots [3 lectures]**
  - Applications to Diophantine equations.
  - Primitive roots and the structure of the group of units modulo n.
- **Quadratic residues [3 lectures]**
  - The group of quadratic residues modulo n.
  - Legendre symbols and Quadratic reciprocity.
- **Sums of squares [2 lectures]**
  - Fermat s ‘two squares’ theorem.
  - Lagrange’s ‘four squares’ theorem.

3.5 MATH36061 – Convex Optimization

Credit Rating: 10

Aims. The course aims to introduce students to modern convex optimization and its applications in fields such as machine learning. The course is designed to cover practical modelling aspects, algorithm analysis and design, and the theoretical foundations of the subject.

Overview. Optimization is the art of optimal decision making under constraints. Convex optimization refers to a set of problems and methods that can be formulated using convex functions and sets; countless problems from science, engineering and statistics can be cast as convex optimization problems and solved using efficient algorithms. The course is intended as an introduction to convex optimization, focussing on the theory, the modelling techniques, and the algorithm analysis and design. Recent developments such as convex regularization and compressed sensing will be discussed. The problem sessions will be used to present applications from machine learning, signal processing, and finance.

Learning outcomes.

On completion of the course, students should be able to:

1. recognize and formulate convex optimization problems as they arise in practise;
2. know a range of algorithms for solving linear, quadratic and semidefinite programming problems, and evaluate their performance;
3. understand the theoretical foundations and be able to use it to characterise optimal solutions to optimization problems;
4. appreciate the role of convex optimization in machine learning, signal processing, compressed sensing, and finance;
5. use the software CVX to model and solve convex optimization problems.

Syllabus.

The lectures and problem sessions will cover the following topics:

1. Overview and examples of optimization problems;
2. Least-squares, gradient descent, Newton's method;
3. Introduction to CVX;
4. Fundamentals of convex analysis and geometry: convex sets and functions, subdifferential calculus;
5. Linear and quadratic programming, semidefinite programming, conic optimization;
6. Optimality conditions, duality theory, theorems of alternative;
7. Interior-point methods, augmented Lagrangians, alternating direction method of multipliers;
8. Applications in machine learning: convex regularization, compressed sensing and matrix completion.

3.6 MATH36001 – Matrix Analysis

Credit Rating: 10

Aims. To introduce students to matrix analysis through the development of essential tools such as the Jordan canonical form, Perron-Frobenius theory, the singular value decomposition, and matrix functions.

Overview. This course unit is an introduction to matrix analysis, covering both classical and more recent results that are useful in applying matrix algebra to practical problems. In particular it treats eigenvalues and singular values, matrix factorizations, function of matrices, and structured matrices. It builds on the first year linear algebra course. Apart from being used in many areas of mathematics, Matrix Analysis has broad applications in fields such as engineering, physics, statistics, econometrics and in modern application areas such as data mining and pattern recognition. Examples from some of these areas will be used to illustrate and motivate some of the theorems developed in the course.

Learning outcomes. On successful completion of this course unit students will be familiar enough with matrix analysis and linear algebra that they can effectively use the tools and ideas of these fundamental subjects in a variety of applications, understand the importance of spectral decomposition, Schur decomposition, Jordan canonical form and singular value decomposition, understand the role of matrix functions in solving differential and algebraic equations, understand how to exploit the structure in certain classes of matrices.

Syllabus.

- **Basics:** Summary/recap of basic concepts from linear algebra including matrices and vectors, determinants, singularity of matrices, rank. [2 lectures]
- **Theory of eigensystems:** Eigenvalues, eigenvectors, and invariant subspaces; reduction of square matrices to simpler form including the Schur decomposition, spectral decomposition for normal matrices and the Jordan canonical decomposition; minimal and characteristic polynomials, Cayley-Hamilton Theorem; Sylvester s inertia theorem. [6]
- **Norms:** Vector norms and matrix norms, bounds for eigenvalues, Gershgorin theorem. [2]
- **Singular value decomposition (SVD):**

  Projectors; pseudo-inverses; application to linear least squares; polar decomposition. [3]
- **Nonnegative matrices and related results:** Irreducible matrices; Perron-Frobenius theorem; diagonally dominant matrices. [2]
- **Matrix functions:** Definitions; the matrix exponential function and application to the solution of differential equations and higher order equations; difference equations and matrix powers. [5]
- **Kronecker product.** Definition, properties and application to the solution of Sylvester s equation (if time). [2]